Epistemic Portrait Envelopes: Partial Identification and Decision Bounds for Scientific Knowledge Integration

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Abstract

Cross-domain scientific synthesis usually underdetermines a unique global account: sources expose different referents, constructs, operationalizations, contexts, modalities and provenance, and extraction may hide the coordinate that would decide whether two claims are identical. We introduce the *epistemic portrait envelope*, the set of global scientific portraits compatible with the observed source-local projections and their licensed mapping constraints. This changes integration from selecting one canonical graph to three separable questions: which claims are point identified, which additional observation would shrink the envelope, and which downstream action is justified while identification remains partial. Five formal results give the framework its scope. A fibre criterion proves that a query is identified exactly when it is constant across all worlds yielding the same observation; otherwise no observation-only rule can be uniformly correct. An information- refinement theorem proves that retaining more truthful source information can only shrink query sets and weakly lower the minimax decision floor. A robust decision bound states when merge, separation or explicit deferral is optimal under asymmetric false-merge, false-split and deferral costs. A joint- completion result shows why pairwise-compatible mappings need not compose into a coherent global portrait. Glue, typed obstruction, plural view and unresolved state become special cases of this set-valued semantics rather than unrelated output labels. A universal factorisation theorem further shows that the identified-set map is the coarsest sufficient claim interface: any other sufficient product must refine it, and information-equivalent implementations must tie under the same loss.

Existing evidence tests slices of the theory without being promoted to end-to-end authority. In a disjoint, prospectively execution-frozen 32-case public-reference holdout, coordinate-governed mapping made zero false merges against 0.1875 for flat predicate canonicalization; the paired difference was -0.1875 (95% bootstrap CI $[-0.34375, -0.0625]$), while its false-split difference from an exact-coordinate conservative control was 0.000 with interval $[0.000, 0.000]$. A separate 400-contract exact battery isolated the scientific-identity decision after strong semantic alignment: the claim-relative layer decided 400/400 exactly versus 250/400 for a strong semantic product, while an information-equivalent typed product tied at 400/400. Most importantly, an archived partial-observation study found nine two-case observationally indistinguishable orbits in a 36-case constructed corpus, imposing a 27/36 ceiling on every rule reading only the released projections. That adverse result motivates the envelope and its floor-adjusted successor endpoint; it is not relabelled as a positive external result. Naturalistic envelope validity, independent multi-domain gold, raw-text robustness and downstream decision gains remain unexecuted.

Keywords: scientific knowledge integration; partial identification; robust decision theory; provenance; schema alignment; scientific pluralism.

1 Introduction

Scientific claims are distributed across disciplines, each with its own referents, constructs, measurements, contexts, and representational choices. A global synthesis can fail even when every local statement is individually well formed. Identical names may denote different objects; nominally similar constructs may rest on incompatible operationalizations; apparent contradictions may differ only in context, modality or attribution; and an extractor may omit the one coordinate that would distinguish a valid merge from a false one.

Most integration pipelines nevertheless seek a single completed artifact: a canonical schema, fused record, knowledge graph, or generated synthesis. That target conflates two questions. The first is epistemic: what do the available source projections identify? The second is decisional: what may a particular downstream user do given what remains unidentified? A more powerful model or a stricter threshold can change the answer to the second question, but it cannot recover a fact that is absent from every input while observationally equivalent source worlds require different answers.

This paper proposes a general alternative: the *epistemic portrait envelope*. Each source remains a provenance-bound local projection. Typed mapping constraints define the set of global portraits jointly compatible with those projections. A query is licensed only when it is invariant across that set; otherwise the system returns the identified set or an unresolved state. A downstream action is selected separately under an explicit loss for false merge, false split and deferral. This formulation absorbs structured extraction, scientific-variable semantics, ontology and schema alignment, provenance-aware integration, pluralism, partial identification and robust decision theory as donor machinery. Its residual contribution is their composition at the scientific-identity interface.

1.1 Contributions

The paper makes five theoretical contributions and supplies three distinct evidence layers.

1. It defines an epistemic portrait envelope over provenance-bound source-local projections and makes identification claim-relative. Scientific plurality within a portrait is distinguished from uncertainty across feasible portraits.

2. It proves a fibre criterion: a query is point identified exactly when it is constant across every admissible world that yields the same observed projections. If two such worlds disagree, no deterministic observation-only rule can be uniformly correct.

3. It proves a universal factorisation result: every query-sufficient integration interface must refine the identified-set map, so products with equivalent information and loss are required to tie.

4. It proves monotonicity under information refinement and defines the observation floor and floor-adjusted avoidable harm. These quantities distinguish an algorithmic failure from irreducible loss caused by missing information.

5. It derives downstream merge/separate/defer bounds under asymmetric costs and shows that pairwise-compatible mapping constraints need not possess a joint global completion.

The first evidence layer is a public-reference mapping study over already-structured projections. A disjoint, prospectively execution-frozen 32-case holdout produced zero false merges for coordinate-governed mapping against 0.1875 for flat predicate canonicalization, while satisfying the declared false-split guard against an exact-coordinate conservative control. The second is a 400-contract exact scientific-identity battery: a claim-relative layer outperformed a strong semantic product lacking the same authority rules, but tied an ideal product given identical information and rules. That tie is required by the theory and prevents an inherent-expressivity claim.

The third layer is an adverse partial-observation result. Mechanical redactions opened an over-resolution failure channel, and a separate 36-case constructed source-record corpus exhibited nine observationally indistinguishable two-case orbits with different gold relations. Any rule reading only the released projections must lose at least one answer per orbit. The current rule reaches the resulting 27/36 ceiling by making eight false merges and one false split; a decisiveness-aware repair removes those errors only by abstaining on both members of each ambiguous orbit and losing the nine answers the current rule happened to get right. We preserve that failure. Its scientific value is to identify the observation floor and to show why the next experiment must measure avoidable harm above it rather than demand an impossible zero.

1.2 Scope and structure

The formal results apply to any scientific integration problem satisfying their stated world, observation, constraint and loss assumptions. The empirical numbers do not inherit that generality. They concern two small public-reference mapping sets, exact authored contracts, and a constructed partial-observation corpus. Naturalistic multi-domain envelopes, raw-text attacks, independent protected gold, and downstream decision improvement remain prospective.

Section 2 positions the framework among the donor literatures. Section 3 defines source-local projections and typed mappings. Section 4 develops the envelope, impossibility theorem, information order, decision bounds and global-composition result. Sections 5–8 report the existing evidence without pooling its authorities. Sections 9 and 10 state the remaining scientific programme.

2 Related work

The framework sits downstream of most machinery that cross-domain scientific integration needs. Extraction, schema matching, entity resolution, fusion, contradiction and stance analysis, and literature-based discovery are treated as available capability. It also imports partial identification and robust decision theory rather than claiming either. The residual question is their composition: after heterogeneous sources have been structured and aligned, which global claims are invariant across all source-compatible completions, which differences certify obstruction or plurality, and what downstream action is justified when scientific identity remains partially identified?

2.1 Partial identification, imprecise probability, and decisions

Partial-identification theory replaces an unjustified point estimate with the set of parameter values compatible with observations and assumptions [20, 23]. Imprecise-probability and robust Bayesian traditions similarly represent a set of probability models and choose actions against the resulting range of expected losses [28, 3]. Blackwell's comparison of experiments orders information structures by the decisions they support [6]. These are direct conceptual donors. The epistemic portrait envelope transports their logic from parameters and experiments to provenance-bound, typed scientific representations: the identified object is a set of global portraits or claim values; an observation refinement shrinks that set; and a downstream loss is declared separately from the scientific relation being represented.

The distinction matters because abstention is not automatically a virtue. It is optimal only under a loss model and an identified set that make deferral cheaper than the worst supported merge or split. Conversely, a determinate mapping is not licensed merely because a model is confident. Point identification is a property of the observation fibre and assumptions, not of the sharpness of a model's posterior. The new contribution is thus not set-valued inference in general, but its composition with scientific coordinate preservation, obstruction, provenance and global portrait construction.

2.2 Constraint composition and global consistency

Ontology and schema matchers commonly propose pairwise correspondences, while constraint-processing research distinguishes local consistency from the existence of a joint solution [9]. The envelope uses that distinction as an authority boundary. A candidate mapping contributes a constraint on feasible worlds; it does not authorize a global identity until all accepted constraints possess a common completion and the claim is stable over those completions. This absorbs generic constraint satisfaction as a donor mechanism while giving its global-consistency result a scientific-identity interpretation.

2.3 Structured scientific knowledge bases and extraction

MUSE [8] constructs a full-text, multi-domain resource of source-grounded problem–solution–rationale structures, and the Discovery Engine [2] distils literature into structured knowledge artifacts and an integrated computational landscape. Both target the representation itself: what a source contains, recovered at scale. The decision studied here begins after that representation exists, and asks whether referent, construct, measurement, context, modality and attribution distinctions can govern a downstream mapping outcome and produce an obstruction when compatibility has not been earned. SciER [30] supplies scientific entity and relation annotations for datasets, methods and tasks in full text, which makes information extraction, discourse processing and entity/relation recovery an input to this work rather than part of it.

2.4 Schema-centric approaches

SciSchema.org [10] provides multidisciplinary, expert-reviewed scientific-process schemas with parameters, measurements, procedures and provenance, and is strong prior art for reusable scientific schema design. SCOPE and SCION [14] introduce an auditable benchmark and reference pipeline for evidence-linked schema induction and optional fusion from text, including conservative fusion behaviour. Executable Schema Contracts [16] discover an executable schema from heterogeneous sources and use it to drive provenance-aware construction and multi-source retrieval. Each of these produces a shared structure that a downstream consumer can rely on; none of them is asked whether a particular proposed mapping preserves the coordinates a specific scientific claim depends on. That is the discriminator evaluated here, over already-structured projections and including cases in which refusing a merge is the correct outcome.

2.5 Alignment, integration, and resolution

Ontology matching [11] studies correspondences such as equivalence, subsumption and disjointness between ontology entities, and language-model systems such as LLMATCH [29] add modern schema-preparation, candidate-selection and column-alignment machinery. Surveys of knowledge-graph construction [13] cover heterogeneous-source integration, ontology and schema matching, entity resolution, fusion, provenance and quality assurance, while reviews of scholarly knowledge graphs [24] cover extraction, construction, refinement and use of semantic scholarly infrastructures.

The distinction pursued here is that a mapping decision is bound to several scientific coordinates at once, and that the exact source, provenance and non-preserved-coordinate information is retained with the resulting decision. Entity resolution and fusion are especially useful pressure cases, because a record-level match may be appropriate at exactly the moment a claim-level merge remains unsafe on construct, measurement, context or modality grounds.

2.6 Claim comparison and stance

Stance detection assigns target-relative labels such as support or opposition; surveys document its use in fact-checking and misinformation settings [12]. Apparent scientific disagreement, however, frequently arises from different context, modality, measurement or attribution coordinates rather than from a genuine opposition, and a mapping record that collapses every difference into one generic contradiction label discards the reason. Preserving that reason, rather than detecting stance, is what is evaluated here.

2.7 Scientific-variable semantics and interoperability

The I-ADOPT interoperability framework [19] provides a community-developed semantic decomposition for machine-readable observable and scientific-variable descriptions, and a recent benchmark builds an expert-annotated corpus of more than one hundred scientific variables and reports that large language models still struggle to construct those structured representations accurately [18]. FAIR 2.0 separates terminological, ontological, referential, schema and logical interoperability, and argues that semantic interoperability requires explicit mappings rather than one privileged ontology [26]. Recent provenance work goes further: DEC reads provenance-bearing statements as epistemic stance and preserves source-relative disagreement instead of collapsing every attributed assertion into a single factual core [25]. Structured variable semantics, generic semantic interoperability and provenance-aware pluralism are therefore supplied by this literature. The residual question is claim-relative and downstream of all of it: once a strong semantic integration stack has produced structured candidate mappings, what further conditions license an assertion of *scientific identity*? The decision contract studied here consumes ontology and schema alignment, provenance, plural-view machinery and consistency checks as inputs, and returns a glue, obstruction, plural-view or unresolved outcome from load-bearing construct, measurement, context, provenance-stance and global-composition conditions.

2.8 Measurement equivalence and construct validity

Measurement-equivalence and replication literatures established long ago that nominally identical outcome measures need not be comparable across settings, and that equivalence can require explicit testing [15]. The mechanism here makes measurement and operationalization one explicit coordinate of the mapping record and allows an unresolved measurement relation to block a global merge on its own.

2.9 Literature-based discovery and pluralism

Swanson's literature-based discovery lineage [21, 22] shows how complementary literatures can expose implicit connections without direct citation links. The bounded requirement added here is on the bridge concept itself: it must retain compatible scientific meaning across the two source-local projections, or the bridge stays an obstruction rather than being promoted to a scientific integration. Scientific pluralism has similarly argued that multiple representations may remain legitimate without collapsing into one complete account [17, 7]; the computational question is whether an integration record can carry a plural or obstructed outcome, with exact source lineage, instead of forcing a canonical merged statement.

2.10 Scientific retrieval and cross-domain synthesis

OpenScholar [1] performs retrieval-augmented synthesis over the scientific literature at scale, and BioSage [27] combines retrieval, specialized agents, cross-disciplinary translation and reasoning for scientific tasks. Retrieval, synthesis, agent orchestration and cross-domain translation are baseline capability for the problem, as are language-model-assisted schema matching and induction. The experiment reported here deliberately holds all of that fixed and varies only the decision rule applied after scientific projections are already structured, so that the construct-validity question can be asked without a retrieval confound.

The general claim is theoretical rather than an empirical promotion: once the donor mechanisms leave more than one scientifically admissible completion, a single canonical portrait requires either additional information, an explicit assumption, or a declared decision rule. The current external evidence is much narrower. On a disjoint, prospectively frozen public-reference holdout of already-structured projections, the typed mapping calculus reduced false merges against flat predicate canonicalization while satisfying the false-split guard declared in advance. Raw-text extraction, naturalistic envelope validity, independent expert construct validity, recoverability and downstream utility require separate evidence.

3 Method

This section defines the scientific mapping objects the public-reference evaluation is defined over. The definitions are data-level: they do not depend on a particular language model or retrieval backend.

The implementation evaluated here is the ORION knowledge-integration layer; it is the artifact under test, and the rest of the paper describes the mapping rule rather than the product, so that each claim reads as a claim about the rule. Its source, protocols and evidence archives are listed in Section 11.

3.1 Three-valued outcomes

Every mapping decision measured in this paper is three-valued rather than binary. A proposed mapping may be determined admissible, determined inadmissible, or left *unresolved*; a coordinate relation may be determined compatible, determined incompatible, or left unresolved. An outcome that could not be determined is recorded and counted separately from an outcome determined to be false, and the two are never merged. The distinction is load-bearing rather than decorative: missing source authority, an absent coordinate or an unsupported relation produces an unresolved outcome, and an unresolved outcome is not evidence that the mapping is wrong. Throughout the paper, an outcome described as undetermined is this third value, carried in the evidence record and in every downstream count, never silently resolved to a negative.

3.2 Source and source-local projection

Let a source record s bind an external document or structured source item to an exact identity, version or revision, locator, and content commitment. The relevant scientific meaning is represented as a source-local projection

$$P_s = \langle R_s, C_s, M_s, X_s, \rho_s, \mu_s, a_s, \delta_s, \Pi_s \rangle, \tag{1}$$

with coordinates for referent R_s, construct C_s, measurement/operationalization M_s, context X_s, polarity ρ_s, modality μ_s, attribution a_s, discourse relation δ_s, and assumptions/unresolved information Π_s.

Each coordinate is a typed hypothesis supported by the source authority available for that case, not a value filled merely to complete a schema. Its observation status is part of the projection: an explicit source statement, an explicit source-level absence, and a coordinate never observed by the extraction process are not interchangeable. In the public-reference datasets, unsupported coordinates remain absent or unresolved; the study does not ask a model to infer missing gold coordinates from raw papers.

3.3 Coordinate-wise comparison

A comparison between projections P_a and P_b is a vector of typed coordinate relations rather than a scalar similarity score:

$$\begin{aligned} q(P_a, P_b) = \langle &\mathrm{ref}(R_a, R_b), \mathrm{const}(C_a, C_b), \mathrm{meas}(M_a, M_b), \mathrm{ctx}(X_a, X_b), \\ &\mathrm{pol}(\rho_a, \rho_b), \mathrm{mod}(\mu_a, \mu_b), \mathrm{attr}(a_a, a_b), \mathrm{disc}(\delta_a, \delta_b) \rangle. \end{aligned} \tag{2}$$

Relations can express compatibility, incompatibility, conditional transformability, or unresolved evidence according to the frozen semantic taxonomies. A proposed relation does not acquire authority merely from lexical or embedding similarity.

3.4 Typed mapping and preservation conditions

A typed mapping m states which coordinate relations it licenses and under what preservation conditions. Its record contains:

- the proposed relation (for example identity, equivalence, conditional transform, subsumption, association-only, no-licensed-mapping, or unresolved);
- the coordinates the mapping claims to preserve;
- conditions that must hold for the mapping to be used; and
- coordinates or information that are not preserved and therefore may not be silently transferred into a global statement.

This representation is deliberately stricter than a successful match score: a mapping can be structurally plausible and still be inadmissible for the scientific claim if a load-bearing condition is absent.

3.5 Gluing, obstruction, and unresolved mapping

Two or more projections may be *glued* into one global statement only when the coordinate relations the mapping requires are compatible and its preservation conditions are satisfied. If a required relation is incompatible, the correct output is a typed *obstruction* naming the failing coordinate or condition. If the available evidence is insufficient, the relation stays unresolved rather than being forced to merge or split. Multiple locally valid but mutually incompatible views may remain a *plural view.*

The confirmatory experiment tests this decision semantics directly. Its strongest discrimination is on polarity, modality, attribution and context cases, where flat predicate canonicalization merges claims that the typed rule keeps separate. It does not identify the necessity of every coordinate; coordinate necessity requires targeted cases that the current datasets do not contain.

3.6 Provenance and recoverable records

Every case retains the external source identity and the frozen record used to derive the expected mapping decision. A mapping result retains the source records and the coordinate conditions supporting its terminal. That provenance is what makes the reported result independently replayable.

The wider design goes further and defines a global portrait

$$\mathcal{G} = \langle \{P_i\}, \mathcal{M}, \sigma \rangle, \tag{3}$$

where $\mathcal{M}$ stores accepted and rejected mappings and σ maps a global statement back to source projections and conditions. Whether a reader can in practice recover a generated portrait this way is a separate empirical question and is not answered by the current mapping study.

3.7 Search-universe feedback is outside the evaluated claim

The wider design also allows newly absorbed representations to change the search universe W and reopen future acquisition routes. That feedback is not manipulated or measured in the public-reference mapping experiment. No result in this paper is attributed to a "W effect," and no expansion of W is used as an ablation supporting the reported mapping result.

3.8 What is and is not claimed

Nothing here claims the first scientific knowledge graph, cross-domain retrieval-augmented system, schema matcher, provenance graph, entity resolver or literature-discovery method. Section 4 makes a general formal claim about what any observation and constraint set identifies, and about decisions under an explicitly declared loss. The supported external empirical object is narrower: over the frozen public-reference mapping cases, explicit scientific-coordinate conditions plus an obstruction state reduce false merges relative to flat predicate canonicalization while satisfying a conservative false-split guard.

Local exact worlds continue to falsify implementation semantics in controlled cases, in the manner established for source-local reconstruction and for typed contradiction detection [4, 5]. They can also witness a formal non-identification result when two admissible worlds share an observation but require different query answers. They do not establish external raw-text extraction quality, recoverability, downstream utility, or broad cross-domain empirical generality; those remain separate prospective questions.

3.9 Method-structure projection

Scientific meaning coordinates describe what a source claims. Structural learning downstream of that also needs a source-local account of *how the source's method transforms its problem*, so the representation is extended with a versioned method-structure projection. It is derived from a provenance-bound method-realization record of the kind produced by source-local epistemic reconstruction [4], and retains exact source, document, version and span identity, the target role and initial obstruction, assumptions and preconditions, representation changes and auxiliary objects where the source supports them, the transformation graph, invariants, progress and termination semantics, reconstruction to the original target, failure and negative-result statements, explicit author rationale when the source actually supplies it, and coordinates typed as unknown otherwise.

This projection stays source-local. In particular, the absence of an author rationale is not permission to invent one. The projection stores the exact realization and span digests that licensed each structured interpretation, and carries no authority to declare that two methods belong to the same formal family.

3.10 Cross-domain method alignment and plural views

Given two source-local projections, a claim-relative mapping μ may be proposed by comparing declared coordinates such as preconditions, assumptions, invariants, progress, termination and reconstruction. The alignment outcome is one of aligned, related, obstructed, or unresolved. The record preserves both projection identities, source-local names and meanings, unmatched assumptions, coordinate-level support and obstruction, and the exact reasons for refusing a merge. An aligned outcome means only that the declared source-local coordinates agree for the proposed mapping purpose; establishing formal equivalence is outside this mechanism's authority.

A clean vocabulary-changing example is numerical bisection against monotone threshold calibration. Their operation names differ, but under the narrow purpose "bounded monotone localization" both require an ordered domain and a bracketed monotone target, preserve a target-containing bracket, make progress by shrinking the admissible interval, terminate at a tolerance, and reconstruct a representative in the original target space. A candidate alignment may therefore be recorded while each donor's vocabulary and lineage is preserved.

A false-merge control uses the same-looking midpoint-and-discard-half surface sequence on a non-monotone response. The monotonicity and bracketing contract no longer holds and the invariant can fail, so the correct result is an obstruction; topology or lexical similarity cannot override that coordinate. A third control uses an opaque recovery procedure whose source does not establish progress or reconstruction semantics. Even when its known effects resemble a rollback method, the alignment stays unresolved rather than filling the missing coordinates from analogy.

Plural views are therefore first-class outcomes: structurally suggestive sources need not collapse into one global method. An obstruction certificate records exactly which coordinates prevent a merge and keeps both source projections recoverable.

3.11 Downstream structural-learning boundary

What this mechanism owns is source-local method projection and candidate plural alignment. Claim-relative structural reduction, substitution and method-family membership are outside it, as is any learned distribution over these versioned objects and any generated method structure proposed from such a distribution. Neither learned similarity nor generated structure changes a source projection or confers scientific authority on it. The bounded pilot reported in Section 6.7 tests only representation and alignment non-vacuity; it is not evidence of general extraction from arbitrary papers.

4 Epistemic portrait envelope theory

The mapping calculus above can be stated more generally. Scientific sources rarely reveal a unique global state. They reveal projections of that state, with different coordinates observed, suppressed, coarsened, or measured under different operationalizations. The appropriate integration object is therefore not always one completed graph. It is an envelope of the global portraits still compatible with the observations and their source-bound constraints. This section develops that object and separates three questions: what the sources identify, what additional information could identify, and what a downstream decision may safely do before identification is complete.

The construction imports the logic of partial identification [20, 23], comparison of information [6], and robust decision theory [3, 28]. The contribution is not a new claim to those donor theories. It is their claim-relative composition with provenance-bound scientific projections, typed mapping constraints, and global-portrait construction.

4.1 Worlds, observation fibres, and portrait envelopes

Let Ω be a set of scientifically admissible worlds. A world $\omega \in \Omega$ fixes the source records, the source-to-projection relations, and a global portrait $G(\omega) \in \mathcal{G}$. The observation operator

$$O : \Omega \longrightarrow \mathcal{Y} \tag{4}$$

returns exactly the evidence available to the integration procedure: the observed projection coordinates, explicit missingness, whatever source identity/version information the interface exposes, and the mapping constraints licensed by those sources. It does not fill an unobserved coordinate from the world that generated it.

The operator is interface-relative. If a procedure receives source text and semantic provenance, those are part of O; if it receives only projections, they are not. Opaque record identifiers that carry no licensed scientific meaning are quotiented by bijective relabelling rather than treated as a lookup key for memorising gold. Every impossibility statement must name this observation boundary.

Definition 1 (Observation fibre and epistemic portrait envelope)**.** For observed evidence $y \in \mathcal{Y}$, define

$$\Omega_y = \{\omega \in \Omega : O(\omega) = y\}, \qquad \mathfrak{E}(y) = \{G(\omega) : \omega \in \Omega_y\}. \tag{5}$$

Ω_y is the observation fibre and $\mathfrak{E}(y)$ is the epistemic portrait envelope. A query $q : \mathcal{G} \to \Theta$ has identified set

$$\Theta_q(y) = \{q(g) : g \in \mathfrak{E}(y)\}. \tag{6}$$

The query is point identified at y exactly when $\Theta_q(y)$ is a singleton.

The query can ask whether two projections may be glued, which typed obstruction holds, what numerical estimand a merged record implies, or which downstream action is warranted. Identification is therefore claim-relative: an envelope can identify one query while leaving another unresolved. A portrait need not be completed in every coordinate before it licenses a specific conclusion.

The envelope also separates scientific plurality from epistemic uncertainty. Plurality is a feature represented *inside* a feasible portrait: several incompatible but source-valid views coexist. Epistemic uncertainty is variation *across* feasible portraits because the available observation does not decide which completion obtains. A plural-view terminal can therefore be point identified; an unresolved terminal cannot.

4.2 The fibre criterion and an impossibility result

Theorem 1 (Fibre criterion for scientific identification)**.** *For any observation y with $\Omega_y \neq \varnothing$ and query q, the following are equivalent:*

1. *q is point identified at y;*
2. *$q \circ G$ is constant on Ω_y;*
3. *there exists an observation-measurable answer $d(y)$ that equals $q(G(\omega))$ for every $\omega \in \Omega_y$.*

If two worlds in the same observation fibre have different query values, no deterministic procedure reading only y can be correct in both worlds.

Proof. The image of Ω_y under $q \circ G$ is exactly $\Theta_q(y)$. It is a singleton if and only if the map is constant on the fibre. In that case its unique value defines $d(y)$. Conversely, if $d(y)$ is correct for every world in the fibre, all those worlds must have query value $d(y)$. The final claim follows immediately: the procedure receives the same input in both worlds and therefore emits the same answer, which cannot equal two different values. □

This theorem turns a partial-observation failure into a scientific result rather than an invitation to invent a fifth rule over the same information. When a coordinate erased by extraction is the only fact distinguishing two source relations, model capacity, prompt changes, and a more elaborate comparison rule cannot restore it. The remedy must either acquire information, declare an assumption that removes worlds from the fibre, or return a set-valued/unresolved answer.

4.3 A canonical sufficient interface

The envelope is not a demand that every integration system share one internal architecture. It defines the least claim-relative interface a downstream consumer must be able to recover.

Definition 2 (Envelope sufficiency)**.** A summary $S : \mathcal{Y} \to \mathcal{Z}$ is sufficient for query q when there exists a decoder h such that

$$\Theta_q(y) = h(S(y)) \qquad \text{for every } y \in \mathcal{Y}. \tag{7}$$

Thus equal summaries may never hide different identified sets.

Theorem 2 (Universal factorisation of sufficient interfaces)**.** *The map $T_q : y \mapsto \Theta_q(y)$ is sufficient for q. Every other sufficient summary S refines it: there is a map h with $T_q = h \circ S$. Consequently, all sufficient implementations presented with the same downstream loss have the same robust loss table and the same set of minimax actions, up to ties.*

Proof. T_q is decoded by the identity map. The factorisation for another sufficient S is its defining decoder. Robust upper loss for action a is a function only of the recovered set, $\sup_{\theta \in T_q(y)} L(a, \theta)$. Equal recovered sets therefore give the same value for every action, and minimisation gives the same optimal-action set. □

This theorem explains both the ambition and the limit of the architecture. A semantic product that exposes exactly the same identified set is not an inferior baseline; it is an information-equivalent implementation and must tie at this interface. A product that collapses two observations with different identified sets is insufficient for that query, no matter how rich its remaining representation is. Any empirical advantage must therefore come from a genuinely finer observation, justified constraints, a more faithful approximation to the envelope, or a different declared downstream loss—not from the name or central location of the layer.

4.4 Information refinement and the observation floor

Say that O_2 refines O_1 when there is a map r such that $O_1 = r \circ O_2$. Thus O_2 retains at least as much information as O_1; the relation is deterministic and makes no probabilistic sufficiency claim. Let $\mathcal{A}$ be a finite downstream action set and $L : \mathcal{A} \times \Theta \to [0, \infty)$ a declared loss. Define the robust decision floor

$$B_q(y) = \min_{a \in \mathcal{A}} \sup_{\theta \in \Theta_q(y)} L(a, \theta). \tag{8}$$

Theorem 3 (Information-refinement monotonicity)**.** *If O_2 refines O_1, then for every realised world ω,*

$$\Theta_q^{(2)}(O_2(\omega)) \subseteq \Theta_q^{(1)}(O_1(\omega)) \quad \textit{and} \quad B_q^{(2)}(O_2(\omega)) \leq B_q^{(1)}(O_1(\omega)). \tag{9}$$

Proof. Every world having the refined observation $O_2(\omega)$ also has coarse observation $r(O_2(\omega)) = O_1(\omega)$, so the refined fibre is a subset of the coarse fibre. Taking images under $q \circ G$ preserves inclusion. For any fixed action, the supremum of its loss over a subset cannot increase; taking the minimum over actions preserves the inequality. □

The quantity $B_q(y)$ is an observation floor, not a defect score for a particular algorithm. It can fall when truthful extraction, measurement, provenance, or source acquisition refines the observation. It cannot be made to disappear merely by changing the decision rule over the same fibre. For a candidate rule δ, its floor-adjusted avoidable harm is

$$X_\delta(y) = \sup_{\theta \in \Theta_q(y)} L(\delta(y), \theta) - B_q(y) \geq 0. \tag{10}$$

This is the quantity a successor experiment should compare. Counting total error without subtracting or stratifying by the observation floor confounds avoidable decision failure with information that no compared rule received.

4.5 Closed-form merge, split, and unresolved bounds

For a binary mapping query, let $\theta = 1$ mean that glue is licensed and $\theta = 0$ mean that it is not. The downstream actions are merge M, keep separate S, and unresolved U. A false merge costs $c_{\mathrm{FM}} > 0$, a false split costs $c_{\mathrm{FS}} > 0$, and deferral costs $c_U \geq 0$. Suppose an independently justified credal description gives only

$$\Pr(\theta = 1 \mid y) \in [\underline{p}(y), \overline{p}(y)]. \tag{11}$$

No point probability is manufactured when this interval is wide.

Proposition 1 (Robust downstream decision bound)**.** *The worst-case expected losses of the three actions are*

$$\overline{R}(M \mid y) = c_{\mathrm{FM}}(1 - \underline{p}), \quad \overline{R}(S \mid y) = c_{\mathrm{FS}}\overline{p}, \quad \overline{R}(U \mid y) = c_U. \tag{12}$$

Consequently, unresolved is minimax-optimal whenever

$$c_U \leq \min\{c_{\mathrm{FM}}(1 - \underline{p}), c_{\mathrm{FS}}\overline{p}\}. \tag{13}$$

With no probabilistic authority beyond the vacuous interval $[0, 1]$, this reduces to

$$c_U \leq \min\{c_{\mathrm{FM}}, c_{\mathrm{FS}}\}.$$

Proof. Merging incurs loss only when $\theta = 0$, so its expected loss is $c_{\mathrm{FM}}(1-p)$ and is maximised at $p = \underline{p}$. Keeping separate incurs loss only when $\theta = 1$, so its expected loss is $c_{\mathrm{FS}}p$ and is maximised at $p = \overline{p}$. Deferral has constant loss. Comparing the three upper risks gives the criterion. $\square$

The costs are downstream and use-specific. A clinical synthesis may assign a larger cost to a false merge than a discovery interface; a screening system may assign a larger cost to a false split. The portrait mechanism should expose the identified set and provenance needed to evaluate those choices, not silently encode one universal utility function.

4.6 Global composition is a joint-completion problem

Let each accepted source or mapping record impose a set $C_j \subseteq \Omega$ of worlds consistent with its licensed conditions. The jointly feasible set is

$$F(y) = O^{-1}(y) \cap \bigcap_{j=1}^{m} C_j. \tag{14}$$

A global glue requires $F(y) \neq \varnothing$; a claim additionally requires its query to be constant on $F(y)$. Pairwise mapping success does not imply either condition.

Proposition 2 (Pairwise compatibility is insufficient)**.** *There exist three mapping constraints that are pairwise jointly satisfiable but have no common global completion. Hence a system that accepts every pairwise-compatible edge can license an inconsistent global portrait.*

Proof. Take $\Omega = \{1, 2, 3\}$ and $C_1 = \{1, 2\}$, $C_2 = \{2, 3\}$, $C_3 = \{1, 3\}$. Every pairwise intersection is nonempty, but $C_1 \cap C_2 \cap C_3 = \varnothing$. $\square$

Constraint processing and acyclic join theory provide stronger algorithms for deciding global consistency than this elementary counterexample [9]. The point here is architectural: pairwise ontology or schema correspondences are proposals for C_j, not self-executing authority for a global merge. Empty joint completion is an identified global obstruction. Multiple completions with different query values are partial identification. Multiple source-valid views inside every completion are an identified plural portrait.

4.7 Evaluation implied by the theory

The envelope theory changes the scientific target from forced-label accuracy to four jointly reported quantities:

1. *validity*: whether protected source-grounded truth is excluded from the returned envelope;
2. *sharpness*: the size or decision diameter of the envelope, reported only together with validity;
3. *floor-adjusted decision harm*: Equation (10) under a declared downstream loss; and
4. *information value*: the reduction in the identified set or decision floor from a specified observation refinement.

These targets envelop the original glue-or-obstruction calculus. A singleton envelope recovers an ordinary determinate mapping; a non-singleton envelope explains why unresolved is required; an empty joint-completion set certifies a global obstruction. The current experiments cover only special cases of this theory. They do not by themselves establish calibrated naturalistic envelopes or downstream utility.

Confirmatory case family	**Cases**
Different name / same referent	13
Polarity, modality, attribution, or context	13
Valid / invalid representation mapping	6
Total	**32**

Table 1: Case-family composition of the disjoint confirmatory public-reference holdout.

5 Public-reference mapping datasets

The empirical result in this paper uses two content-addressed *public-reference mapping datasets*. They contain already-structured scientific projections assembled from pinned external human and expert resources and from deterministic standards. They are not newly commissioned expert gold, and they do not evaluate raw-text extraction.

5.1 Initial public-reference set

The initial mapping study contains 32 cases generated from pinned revisions of MUSE, SciFact, and SciSchema. The portable gold archive is bound by content digest, and the evidence package binds the external-source revisions and evaluator identity while retaining a checksum manifest rather than redistributing restricted third-party text. This first result is treated as development evidence; it is not pooled into the confirmatory verdict.

5.2 Disjoint confirmatory set

A second 32-case holdout was selected by an outcome-blind deterministic round-robin window over the sorted source pools, starting after the initial window and excluding all initial case identifiers. Its overlap with the first set is zero. Before any confirmatory system output was evaluated, the execution manifest froze the portable gold digest, the MUSE, SciFact and SciSchema revisions, evaluator component identities, bootstrap seed, margins, and terminal rule. Every digest named in that freeze is listed in Section 11.

Table 1 gives the composition. The 13 different-name/same-referent cases come from the MUSE cross-domain stratum. The 13 polarity/modality/attribution/context cases come from a scientific-claim-verification stratum. The six representation-mapping cases span materials, physics, and psychology and use deterministic standards where appropriate. Each record retains its source authority class and exact frozen input identity.

5.3 What the public-reference labels mean

The evaluated decision is a mapping-level terminal over already-structured projections. Depending on the source relation and the tested rule, the correct outcome can permit integration, forbid integration, or preserve an unresolved relation. The study evaluates false merges and false splits under these frozen cases. It does not claim that a model recovered the semantic coordinates from raw papers.

The confirmatory set is particularly discriminating for the family of polarity, modality, attribution and context cases: flat predicate canonicalization false-merges 6 of 13 cases (rate 0.4615), where coordinate-governed mapping makes no false merges. The different-name/same-referent and representation-mapping strata are clean controls on this holdout: all evaluated rules are correct there. That localization is reported rather than hidden, because the observed pooled effect is not evidence that every coordinate is necessary.

5.4 The broader expert-atlas design is not the dataset used here

A prospective expert-atlas design is preserved alongside this study: a protocol, annotation schema, handbook, adjudication policy, and 32 seed-to-gold templates spanning a broader eight-family design. Those templates are not expert gold. No independent dual annotation, per-coordinate inter-annotator agreement, specialist adjudication, or prospectively bound raw-text model run exists for that broader object, so its outcome remains undetermined and it is excluded from the claim made here rather than being presented retrospectively as the dataset used.

5.5 Data and licensing boundary

The public-reference package stores source identities, pinned revisions, locators, provenance and content commitments needed for replay. Third-party text is not vendored where redistribution is not permitted. A final submission archive will freeze the exact repository subject and stable public release URL for the legally shareable package; the absence of a permanent archive at the current drafting stage does not change the scientific result.

6 Evaluation

The empirical claim in this paper is governed by an execution-frozen protocol that was created after the initial 32-case result but before any confirmatory system output was examined. The predecessor result was allowed to motivate replication; it was not allowed to change the second holdout, the evaluator, the margins, or the pass criteria. The broader raw-text and expert-gold design remains a separate, unexecuted study. Both freezes are identified in Section 11.

6.1 Primary confirmatory hypothesis

The frozen replication hypothesis has two non-compensatory conditions:

False-merge superiority: the coordinate-governed-minus-flat false-merge point estimate must be at most -0.05, and the upper bound of the paired 95% bootstrap interval must be below zero.

False-split non-inferiority: the coordinate-governed-minus-exact-conservative false-split point estimate must be at most $+0.03$, and the upper bound of its paired 95% bootstrap interval must be at most $+0.03$.

The holdout passes only if both conditions hold. It contains 32 cases, has zero overlap with the initial 32-case set, and was bound by exact content digest before any confirmatory outcome was accessed.

6.2 Compared mapping rules

The confirmatory primary comparison uses three deterministic rules over the same already-structured mapping cases:

1. **Typed mapping calculus**: evaluates the scientific coordinates the case requires and can retain an obstruction rather than force a merge.

2. **Flat predicate canonicalization**: treats compatible surface predicates as sufficient merge evidence without the coordinate distinctions. This is the predeclared false-merge comparator.

3. **Exact-coordinate conservative control**: requires exact coordinate compatibility and can abstain where it lacks such a match. This is the predeclared false-split and non-inferiority control.

These are mapping-layer comparators, not attempts to reproduce a full external knowledge-graph, retrieval-augmented or schema-fusion system. Stronger schema-induction and provenance-contract baselines would be useful post-outcome construct-validity pressure, but they cannot rewrite a confirmatory status that was frozen beforehand.

6.3 Secondary ablations

The frozen mapping evaluator also reports descriptive paired ablations. The covered attacks are:

- **force compatibility without obstruction**: removes the explicit non-merge terminal and chooses a compatible mapping even where the coordinates prohibit it;
- **remove modality/polarity/attribution/discourse**: removes the coordinate group carrying the main confirmatory discrimination;
- **remove referent**, **construct**, **measurement**, and **temporal context**: retained as zero-effect controls on this particular holdout.

The ablations are secondary and descriptive. The protocol explicitly forbids turning their intervals into new confirmatory significance claims, and forbids reading zero effect on an under-supported stratum as evidence that a coordinate is unnecessary.

6.4 Metrics and statistics

The confirmatory decision uses case-level false-merge and false-split rates. Wilson 95% intervals describe individual rates. Paired differences use a 10,000-resample percentile bootstrap with frozen seed 20260817. The mapping rules are deterministic, so the protocol uses one execution per case rather than artificial stochastic repeats. Results are reported pooled and descriptively by case family and source stratum. The initial 32 cases are never pooled into the confirmatory verdict.

The primary decision rule is therefore fixed entirely by the two paired comparisons above. No post-outcome margin change, exclusion, reweighting, or replacement baseline can change whether the confirmatory run passed.

6.5 Custody and fail-closed rules

Before any confirmatory output was produced, the execution manifest bound the portable confirmatory gold digest, zero overlap with the predecessor gold, the exact MUSE, SciFact and SciSchema source revisions, the evaluator component identities, the bootstrap seed and pass margins, and a rule that any digest mismatch, source drift, evaluator drift or case overlap leaves the result undetermined rather than silently re-evaluating a different study.

The gold is not an input to the evaluated mapping rule. The evaluator consumes the candidate mapping result and the protected expected decision only after the system output has been sealed.

6.6 Reproducibility

The result is reproduced by deterministic public-reference scripts over immutable archives: the confirmatory analysis, publication figures and tables, source registry, evaluator identities, and an independent replay receipt are all retained, and two make targets regenerate the headline analysis and the publication artifacts from committed inputs. Section 11 names each of them.

This reproducibility claim does not extend to the unexecuted raw-text expert atlas, to full external retrieval or schema baselines, to generated-portrait recoverability, or to downstream answer quality. Each of those remains undetermined until its own prospective run exists.

6.7 Bounded method-structure pilot

The public-reference scientific-meaning experiment is not an evaluation of method-structure extraction, and is not offered as one. The larger expert-annotated method-structure corpus originally proposed for this extension—50–100 papers and algorithms across at least five fields, with independent annotation—was not available for this submission track and is therefore outside the empirical claim.

Instead, a prospectively frozen non-vacuity pilot uses six authored exact-ground-truth method pairs across numerical methods, graph algorithms and transactional workflows. The panel contains three structurally aligned pairs, two explicit obstructions and one source-insufficient case whose correct outcome is unresolved. Eight controlled corruptions delete or change an invariant, a reconstruction map, a dependency order, a precondition, failure semantics, lineage, an authority boundary, or an explicitly unknown progress coordinate. The typed representation and alignment path reproduces all six frozen pair labels and detects all eight material corruptions. A deliberately simple transcript and action-overlap comparator resolves only two of the six pair labels.

That supports a narrow claim only: the typed source-local objects are discriminating on a small exact-ground-truth panel and can preserve an explicit unresolved state. It does not estimate annotation agreement on real papers, false merge or split rates over natural scientific prose, end-to-end extraction quality, or the performance of any learner built downstream of these objects. Those remain undetermined until independently annotated evidence exists, and the pilot narrows the method-projection claim rather than establishing a general cross-domain method-learning result.

7 Post-saturation validation: representational compatibility is not scientific identity

The public-reference result answers a question about flat canonicalization. A stronger comparator makes a wider question testable: if a semantic integration stack has already produced well-structured candidate mappings, does that structure by itself authorize an assertion of scientific identity? After the public-reference result was frozen, a separate study was opened to test that without rewriting the original 32-case evidence. Its protocol was frozen before protected outcomes, and it credits scientific-variable decomposition, ontology and schema alignment, provenance-aware integration, plural and inconsistency representation, and generic consistency checking as prior work. The only disputed interface is whether those representational outputs authorize scientific identity on their own.

The scientific-identity contract battery. The protected battery contains 400 exact already-structured contracts: five scientific domains, eight mapping archetypes and ten disjoint variants per cell. The correct terminal for each contract is to glue, to record an obstruction, to record a plural view, or to leave the mapping unresolved.

Strong products. The primary comparator is granted strong structured semantic information: construct, measurement, context, provenance and explicit missingness. It correctly handles direct construct, measurement and context mismatches and missing load-bearing coordinates. Its systematic differences are three: it treats provenance as trace rather than as scientific stance, it forces a canonical choice for legitimate one-to-many mappings, and it accepts pairwise-compatible mappings without a global scientific-consistency check. A second comparator performs canonical matching and represents the usual merge-on-high-local-match pressure. A third is an ideal fully typed product that receives exactly the same coordinates and rules as the layer under test and is required to tie; it is an expressivity boundary rather than a weak baseline.

Protected result. The claim-relative compatibility layer reached exact scientific-integration decisions on 400/400 protected cases, against 250/400 for the strong semantic product and 50/400 for the canonical matcher. The paired difference against the strong product was +0.375 with a domain-stratified 95% bootstrap interval of $[0.3275, 0.4225]$; McNemar discordance was 150–0. The layer made no false merges, retained 1.0 clean-merge accuracy, and the paired effect was +0.375 in each of the five exact domains. An independent implementation reconstructed all four arms and the canonical protected-row digest exactly.

Boundary result. The ideal typed product also reached 400/400, with zero decision mismatches against the layer under test. The result therefore does not establish inherent expressivity or an advantage from centralization. What it supports is a bounded authority distinction: in the tested exact contracts, representational compatibility is weaker than scientific identity authority, and a claim-relative compatibility layer adds value over a strong semantic integration product by preserving obstruction, plurality and unresolved state where canonical identity is not licensed. The battery does not establish raw-text extraction, live ontology-engine superiority, expert agreement on new corpora, or open-ended cross-domain integration; each of those remains undetermined.

In the envelope theory of Section 4, the ideal-product tie is not merely a negative control. It is an information-equivalence result: two procedures receiving the same observation, admissible-world constraints and decision rule should return the same identified set and terminal. The strong product's 150 disagreements arise from omitted authority conditions in its frozen contract, not from an inability to represent those conditions in principle. The scientific target is therefore a sufficient interface for identification and action, not a permanently privileged implementation.

8 Results

The evidence layers are deliberately kept separate. The public-reference projection-to-mapping protocol has been executed twice: an initial frozen 32-case study, followed by a disjoint, prospectively execution-frozen 32-case confirmation. The broader raw-text expert atlas, generated-portrait recoverability, downstream utility and full coordinate necessity are separate questions whose outcomes remain undetermined; they are not treated as missing pieces of the mapping result reported here. All numbers below come from immutable archived artifacts, listed in Section 11.

8.1 Initial public-reference mapping result

The public-reference builder used pinned external source authorities from MUSE, SciFact, and SciSchema and did not fill unsupported semantic coordinates with model guesses. The first deterministic build produced 32 already-structured mapping cases across five recorded strata and three case families: different-name/same-referent, polarity/modality/attribution/context, and valid/invalid representation mappings. A second freeze pass reproduced the portable gold artifact byte for byte.

On those 32 cases, the coordinate-governed comparison rule classified all expected relations correctly (accuracy 1.000), with false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had accuracy 0.875 and false-merge rate 0.125, with no false splits. The paired false-merge difference was -0.125 with a 95% paired percentile-bootstrap interval of $[-0.250, -0.03125]$ (10,000 resamples; seed 20260817).

An exact-coordinate conservative control also produced zero false merges but abstained on 0.125 of cases and achieved accuracy 0.875. The initial result therefore motivated a prospective replication without itself becoming the final confirmatory authority.

8.2 Prospectively frozen confirmatory replication

A second, disjoint 32-case holdout was selected with the same outcome-blind round-robin construction at selection offset 32, excluding every initial case identifier. It was frozen before any confirmatory output of any compared rule was examined; the execution manifest bound its portable digest, upstream source revisions, evaluator identities, bootstrap seed, margins, and the terminal rule before evaluation.

The rule declared in advance required both (i) a coordinate-governed-minus-flat false-merge difference at most -0.05 with the upper end of its paired 95% bootstrap interval below zero, and (ii) a coordinate-governed-minus-exact-conservative false-split difference no larger than $+0.03$ with the interval upper bound no larger than $+0.03$. The holdout met both conditions. Coordinate-governed mapping had false-merge rate 0.000 and false-split rate 0.000. Flat predicate canonicalization had false-merge rate 0.1875 and accuracy 0.8125. The paired false-merge difference was -0.1875 with 95% interval $[-0.34375, -0.0625]$. The false-split difference against the exact-coordinate control was 0.000 with interval $[0.000, 0.000]$. The confirmatory primary verdict was therefore a **pass** under the rule frozen before those outputs.

The effect is localized to the strata that discriminate the tested rules. Among the 13 polarity/modality/attribution/context cases, flat canonicalization false-merged 6/13 (0.4615), where coordinate-governed mapping false-merged none. The 13 different-name/same-referent cases and the six representation-mapping cases were correct under all compared rules. These strata are reported descriptively and are not pooled with the initial 32 cases to inflate the confirmatory sample size.

8.3 Covered ablations

On the confirmatory holdout, forcing compatibility without the explicit obstruction distinction increased false merges by $+0.1875$ (95% paired bootstrap interval $[+0.0625, +0.34375]$) and reduced accuracy by the same absolute amount. Removing the modality/polarity/attribution/discourse coordinate group produced the same false-merge increase on the covered claim-verification cases.

Removing referent, construct, measurement, or temporal context has zero measured effect on this holdout. Those zero effects are preserved as coverage limitations, not read as proof that the coordinates are unnecessary or dispensable. The confirmatory protocol treats all ablations as descriptive secondary analyses rather than as new significance claims.

8.4 Independent verification and local falsifiers

A separate implementation independently reproduced the confirmatory mapping result and its case-level decisions. The verification is bounded to the public-reference mapping claim; it does not promote the raw-text or downstream questions.

Synthetic exact-world tests separately exercise referent and construct identity, measurement equivalence, context, gluing, obstruction, plural views, preservation conditions, and typed mapping hypotheses. These remain engineering and semantic falsifiers rather than external empirical evidence.

8.5 Partial observation: a preserved adverse result and its bound

A staged partial-observation study asked whether adding an observedness-aware decision rule could repair over-resolution when one side of a mapping pair lost the coordinate that originally carried the decision. Mechanical redactions created 48 probes from the frozen atlases: 12 derivation cases, eight held-out public-reference cases, and 28 held-out exact cases. The current rule returned a determinate merge on all 48 although the redacted observation did not identify that decision. Rules that abstained on every asymmetric absence repaired this channel but destroyed determinate answers on intact cases. A more selective rule abstained only when admissible completions disagreed, improving on blanket abstention but still failing its zero-harm gate.

The last amendment showed why another observation-only rule cannot remove that cost. Its 36-case constructed source-record corpus contains 27 canonical observation orbits. Nine are two-case orbits in which the released projections are identical up to bookkeeping, identifier relabelling and left–right orientation, while the source-record gold relations differ. At least one member of each such orbit is therefore unrecoverable from the projections, so the maximum exact agreement of any candidate-visible deterministic rule is 27/36. The current rule reaches that ceiling with eight false merges and one false split. The selective unresolved rule avoids both error types by abstaining on both members of each ambiguous orbit, thereby losing the nine answers the current rule happens to get right there.

This is an exact identification result on a constructed corpus, not external evidence that natural scientific atlases have the same frequency or cost of missingness. The historical gates remain failed. The scientific consequence is the observation floor formalised in Theorem 3: a successor should compare avoidable false merge or downstream harm above that floor and should seek truthful observation refinements, rather than relabel a zero-harm threshold or claim that a more elaborate rule can recover erased information.

8.6 What this result does not establish

The reported result does not establish:

- **Raw-text end-to-end superiority:** no prospectively bound raw-text extractor or model comparison against strong retrieval-augmented or schema systems supports this paper's headline;
- **Generated-portrait recoverability or downstream utility:** no frozen reader-level or downstream task result is promoted here;
- **Naturalistic envelope calibration:** the constructed partial-observation bound proves an observation-level impossibility on its frozen cases, but does not estimate the prevalence, sharpness or loss of partial identification in external scientific corpora;

- **Universal coordinate necessity:** targeted cases for referent, construct, measurement, temporal context and additional obstruction families do not exist in these datasets;
- **An executed expert eight-family gold:** the broader seed templates and annotation protocol remain prospective rather than being relabelled as the public-reference dataset.

Each of those outcomes therefore remains undetermined, and stays visible as undetermined so that the replicated mapping result cannot be read as authority for the wider design. Any future positive has to arrive as a new prospectively frozen artifact rather than as a widening of the current claim in prose.

9 Limitations and ethics

9.1 Limitations

Theory–evidence separation. The fibre criterion, information-refinement monotonicity, robust decision bound and pairwise-consistency counterexample are mathematical statements under their declared assumptions; their validity does not depend on the sample sizes below. Their scientific usefulness does. The framework does not yet show that a deployed extractor can construct a valid and sharp naturalistic portrait envelope, that its admissible-world model contains the relevant source relation, or that experts will agree on the constraints defining that model. A misspecified envelope can be confidently narrow and wrong. Empirical coverage, sharpness and assumption sensitivity must therefore accompany any future application.

Decision losses are not scientific facts. The merge/separate/defer bound makes costs explicit but does not estimate them. False-merge, false-split and deferral costs depend on the downstream task and stakeholders. A mapping mechanism should expose identified sets and provenance; it should not launder one author's utility function into a universal scientific relation. No downstream cost elicitation or protected decision study has yet been executed.

What is and is not owned here. Almost every mechanism this work depends on is prior art, and the boundary is worth stating once rather than repeatedly. Structured scientific-variable semantics, ontology and schema alignment, provenance modelling, provenance-aware pluralism, evidence-linked schema induction and fusion, entity resolution, stance and contradiction analysis, retrieval-augmented synthesis and generic consistency checking are all donor mechanisms, described and cited in Section 2. The residual is the claim-relative decision that runs after all of them: which coordinate differences license an assertion of scientific identity, and which oblige the record to keep an obstruction, a plural view or an unresolved relation visible instead. A result in favour of that decision layer is not a result in favour of the donor mechanisms, and none of the evidence here should be read as ownership of them.

Scope of the reported external result. The reported external result is the public-reference projection-to-mapping study, not an end-to-end expert atlas. It contains two separately frozen 32-case mapping sets assembled from pinned public human and expert resources; the confirmatory set is disjoint at the case level from the initial set. These cases are already-structured scientific projections. They therefore test mapping and obstruction semantics, not whether a model can reliably extract all required semantic coordinates from raw papers.

Scope of the post-saturation battery. The scientific-identity contract battery is a separate, prospectively frozen exact-contract study, not a pooled extension of the 32-case public-reference result. Its five domain labels denote heterogeneous scientific mapping contracts with mechanically defined ground truth; they are not executions of five live ontology or schema engines, and they do not substitute for newly commissioned expert agreement. The battery supports the bounded decision principle that representational compatibility can be weaker than scientific identity authority in the tested contracts. It does not establish ontology-engine-agnostic superiority or open-ended cross-domain integration.

Expert subjectivity and theory-ladenness. Scientific identity and measurement-equivalence judgments are theory-laden and may depend on disciplinary background. The prospective expert-atlas design mitigates this through a written handbook, independent double annotation, coordinate-wise agreement reporting, domain-expert escalation, and an explicit unresolved state. That expert-gold study has not been executed, so this manuscript reports no inter-annotator agreement and presents no seed template as expert gold.

Domain and case-family coverage. The public-reference route covers only the case families its pinned source authorities support. Its confirmatory effect is concentrated in polarity/modality/attribution/context cases; different-name/same-referent and representation-mapping cases are useful controls but do not discriminate the compared rules on that holdout. The successor battery broadens the exact contract families but still does not establish universal cross-domain adequacy or the necessity of every semantic coordinate. In particular, a coordinate's usefulness in the exact successor does not imply that every scientific domain must encode that coordinate in the same representation.

Evolving terminology. Scientific terminology and operationalization change over time. The mapping study is bound to specific source revisions and does not establish robustness to temporal drift, historical conceptual change, or cross-lingual terminology.

Ontology and schema dependence. The semantic-coordinate representation is a design choice. Other representations may encode additional causal, probabilistic, temporal, or normative structure. The current evidence supports a bounded typed mapping and obstruction result and a bounded claim-relative compatibility successor; it does not establish that the chosen coordinate system is uniquely correct or optimal. An ideal product given the same scientific coordinates and rules ties the successor layer exactly.

Extraction error propagation. The text-to-scientific-meaning extraction layer remains unexecuted. Extraction errors could propagate into mapping, gluing and obstruction decisions. Because no prospectively bound raw-text extractor run supports the headline result, this paper reports no stage-attributed extraction error and claims no end-to-end superiority over raw-text, retrieval-augmented or schema systems.

Recoverability and downstream utility remain open. The design intends global statements to retain source projections, mapping paths, preservation conditions, and non-preserved coordinates. Local tests verify those mechanics, but the current external and exact-contract studies establish neither reader-level recoverability of generated portraits nor downstream scientific-answer improvement. Both remain undetermined pending their own frozen evaluations.

The external partial-identification successor is unexecuted. A distinct prospective successor freezes 768 independent source-artifact-family clusters across 16 strata and four domains, requires nonzero variation in referent, construct, measurement and temporal-context coordinates, and defines its endpoint as avoidable false merge or downstream decision harm above the proven observation floor. Its local preflight does not grant scientific authority. The required source-disjoint external atlas, independent protected gold, raw-text attack corpus and downstream-evaluator custody do not yet exist; no outcome from that successor is reported here.

9.2 Ethics

The public-reference study reuses published research resources and collects no new human-participant data. The successor battery uses exact authored contracts rather than new human annotations. The broader expert-annotation protocol is prospective and has not been executed. An obstruction or a plural view should be presented as a provenance-bound record of incompatibility or of unresolved mapping, not as a judgment that one scientific perspective is intrinsically superior. Source lineage is retained so that a downstream reader can inspect the evidence and the conditions behind an integration decision.

10 Conclusion

A global scientific portrait should not be forced to say more than its sources identify. This paper replaces the single canonical portrait with an epistemic portrait envelope: all global portraits compatible with the observed source-local projections, provenance and licensed mapping constraints. The change is not a retreat to vagueness. It produces exact conditions for when a claim may be sharp and exact bounds for acting when it cannot be.

The fibre criterion proves that a query is point identified precisely when it is constant across every admissible world yielding the same observation. If two worlds share the observation and require different answers, no rule over that observation can be uniformly correct. The universal factorisation theorem identifies the implementation-neutral target: every sufficient interface must refine the query's identified-set map, and information-equivalent products are required to tie. The information-refinement theorem then identifies the constructive route upward: truthful source acquisition, extraction, measurement or provenance can shrink the envelope and weakly lower the minimax decision floor. More elaborate inference over unchanged evidence cannot. A binary robust-decision result states when merge, separation or deferral minimises worst-case loss, and the joint-completion result shows why pairwise-compatible mappings still require a global consistency check.

These results unify four terminals. Glue is licensed when the relevant claim is invariant across the feasible envelope. A typed obstruction is an identified failure of joint compatibility. A plural portrait represents multiple source-valid views inside the scientific object. Unresolved state records variation across feasible completions. In particular, scientific plurality and analyst ignorance are not the same state.

The evidence supports narrower instances. On a disjoint 32-case confirmatory holdout assembled from pinned public human and expert resources, the typed comparison made no false merges, while flat predicate canonicalization false-merged 0.1875 of cases; the paired difference was -0.1875 with a 95% bootstrap interval of $[-0.34375, -0.0625]$. Against the exact-coordinate conservative control, the false-split difference was 0.000 with interval $[0.000, 0.000]$. A separate 400-contract exact battery produced 400/400 exact decisions for the claim-relative layer against 250/400 for a strong semantic product; an ideal product receiving the same coordinates and rules tied at 400/400. The tie rules out inherent expressivity or centralisation as the explanation.

The most informative result is adverse. In the constructed partial-observation corpus, nine two-case orbits share exactly the evidence visible to every compared rule while carrying different source-record gold. No observation-only rule can exceed 27/36 exact answers. The current rule attains that ceiling by accepting eight false merges and one false split; the selective unresolved repair avoids those determinate errors only by giving up the nine answers the current rule gets right in the same orbits. The failed historical gate is preserved. Its successor asks the scientifically stronger question: which system reduces *avoidable* harm above that observation floor after independent, source-disjoint observation and gold are supplied?

The resulting programme is wider than a coordinate mapper but does not borrow authority from unexecuted studies. Partial identification, robust decision theory, scientific-variable semantics, ontology and schema alignment, provenance, constraint processing and pluralism are donor theories. The proposed synthesis turns their outputs into a claim-relative envelope and a downstream decision interface. Naturalistic envelope validity, raw-text robustness, independent multi-domain gold, source recoverability and measured downstream gains remain open. Those are now explicit research problems with identifiable estimands and falsifiers, not reasons to narrow the theory or to rename missing evidence as success.

11 Data and code availability

The implementation repository contains the mapping calculus, the frozen public-reference datasets and their builders, the deterministic evaluator, the successor battery, and the publication-layer regeneration commands. Every artifact named anywhere in this paper is listed here with its path, so that no repository location has to be carried in the narrative. Paths below are relative to the paper directory unless they begin with `src/` or `research/`.

11.1 Preregistration

The end-to-end study was preregistered and frozen before any outcome was accessed, under the identifier `P3.cross-domain-atlas.v1`. Its machine-readable freeze is `protocol/PROTOCOL_V1.json`, with the annotation schema, handbook and adjudication standard in `protocol/ANNOTATION_SCHEMA_V1.json`, `protocol/ANNOTATION_HANDBOOK_V1.md` and `protocol/ADJUDICATION_POLICY_V1.md`. That expert-gold design has not been executed; it is preserved so that a reader can see what was planned, and it supplies no evidence for anything reported here.

The executed mapping study has its own preregistration chain. The design freezes are `protocol/PUBLIC_REFERENCE_PROTOCOL_V1.json` and `protocol/PUBLIC_REFERENCE_PROTOCOL_V1_1.json`, and the execution freeze for the confirmatory holdout is `protocol/PUBLIC_REFERENCE_CONFIRMATORY_EXECUTION_V1.json`, registered under the identifier `P3.public-reference-mapping.v1.1-confirmatory`. A reader checking whether a reported analysis was planned or chosen after the fact should compare against that freeze; it is the reason every number in Section 8 can be traced to a decision made in advance of seeing it. The successor battery is frozen separately under `research/claim_expansion/p3/`, whose protocol, protected freeze, outcome-access receipt and result are `P3_X_PROTOCOL_V1.md`, `P3_X_PROTECTED_FREEZE_V1.json`, `P3_X_OUTCOME_ACCESS_RECEIPT_V1.md` and `P3_X_PROTECTED_RESULT_V1.json`.

11.2 Mechanism implementation

The coordinate comparison, bridge admissibility and mapping semantics defined in Section 3 are implemented in `src/orion/knowledge/semantics.py`; the public-reference builders, evaluator, analysis and publication projection are `src/orion/study/p3_public_reference_build.py`, `src/orion/study/p3_public_reference_holdout.py`, `src/orion/study/p3_public_reference.py`, `src/orion/study/p3_public_reference_analysis.py` and `src/orion/study/p3_public_reference_publication.py`. The independent evaluator used for replay is `src/orion/study/p3_independent_evaluator.py`.

11.3 Frozen public-reference datasets

The initial gold archive is `gold/adjudicated/public-reference-v1/PUBLIC_REFERENCE_GOLD_V1.jsonl` with its freeze manifest `gold/adjudicated/public-reference-v1/PUBLIC_REFERENCE_FREEZE_MANIFEST_V1.json`; its portable SHA-256 is `35f9e39b75ff53b7f0ec82cd03ebcaaa82509ee0aea3f5b96aac3fd62c854ed8`. The disjoint confirmatory holdout is the same pair of files under `gold/adjudicated/public-reference-v1.1-confirmatory/`, with portable SHA-256 `13a76c68c149c2552f3543babeca6e1ad5afe23c45ea9c0dc365c1445cf2782b`. The pinned external source authorities and their revisions are recorded in `gold/PUBLIC_REFERENCE_SOURCE_REGISTRY_V1.json`, and the sampling and authority rules are `gold/PUBLIC_REFERENCE_AUTHORITY_POLICY_V1.md` and `gold/GOLD_METHODOLOGY_V1.md`. Restricted third-party text is not redistributed; the packages retain source identity, version, locator and content commitment instead.

11.4 Frozen results

The initial build report, evaluated summary, analysis, provenance record and checksum manifest are under `evidence/public-reference-v1/`; the external-source revisions and evaluator identity are bound in `evidence/public-reference-v1/PROVENANCE.env`. The confirmatory counterparts are under `evidence/public-reference-v1.1-confirmatory/`: the pre-outcome binding is `EXECUTION_MANIFEST.json`, the reported numbers are `CONFIRMATORY_ANALYSIS.json`, and the publication figures and tables are under `publication/` with their own `SHA256SUMS`. The bootstrap seed 20260817 and both pass margins are recorded in the execution manifest rather than chosen afterwards.

The independent reproduction of the confirmatory mapping result is `research/verification/records/P3.C5.confirmatory-mapping.json`; the successor battery's independent verifier and its record are `research/claim_expansion/p3/verify_p3_x_independent.py` and `research/claim_expansion/p3/P3_X_INDEPENDENT_VERIFICATION_V1.json`. The bounded method-structure pilot of Section 6.7 has its protocol, runner and results in `research/extensions/p1-p3-structure/` and its receipt at `evidence/method-structure/BOUNDED_PILOT_RECEIPT_V1.md`.

The mapping from each result-bearing claim to the archived artifact that supports it is `CLAIM_LEDGER_V1.md`, with the manuscript map at `CLAIM_LEDGER_MANUSCRIPT_MAP_V1.md`.

11.5 Reproduction

Two make targets regenerate the reported analysis and the publication artifacts from committed inputs: `make paper03-public-reference` produces the headline analysis, and `make paper03-public-reference-publication` produces the publication figures and tables. Both are deterministic, and continuous integration re-runs the second on a clean checkout and requires the regenerated bytes to be identical to the committed ones. The short reproduction route is `REPRODUCE.md`.

Reproduction establishes only what the frozen artifacts establish. The unexecuted raw-text expert atlas, external retrieval and schema baselines, generated-portrait recoverability and downstream answer quality have no artifacts here, and their outcomes remain undetermined rather than absent by oversight.

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